

Diagnostics Module — Slice 2

Test Plan

Imaging · Outsourced Labs · Resource Scheduling · Pre-Authorization · Separation of Duties

Tenant	MBS Hospital
Scope	Slice 2 build — extends the OPD Pathology Test Plan (Slice 1)
Prepared for	Functional / UAT Testing
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1. Purpose & Scope

This document is a companion to the earlier *OPD Diagnostics Workflow & Test Plan* (Slice 1 — OPD Pathology). It covers everything added since: **Imaging** (X-ray/CT/MRI/USG — report-only, no image storage/viewer), **Outsourced lab** tests sent to a partner lab, **machine/resource scheduling**, **TPA/insurance pre-authorization** thresholds, a per-tenant **separation-of-duties** toggle, and three smaller workflow additions (pending-recommendation follow-up, order payment top-up, and Reception's visibility into the Collection worklist).

Every test case below was verified against a live running instance before this document was written — none of it is aspirational.

2. Test Accounts

ROLE	USERNAME	PASSWORD	USED FOR
Admin	ADMIN-001	Admin@12345	Catalogue, resources, pre-auth config, MFA required
Reception	MBSRECP2026001	Test@12345	Booking, payment, follow-up worklist
Radiographer	MBSRADIOG001	Test@12345	Performing imaging studies
Radiologist	MBSRADIOL001	Test@12345	Authoring findings
Radiologist (2nd)	MBSRADIOL002	Test@12345	Verifying — must be a <i>different</i> login from whoever authored
Lab Technician	MBSLABTECH001	Test@12345	Outsourced dispatch, returns, result entry

All seeded, tenant-scoped test fixtures on the MBS Hospital tenant — not production credentials. The two Radiologist accounts are required together to test the separation-of-duties rule properly.

3. Workflow Overviews

3.1 Imaging

- 1

Doctor recommends a study
DOCTOR
Same Investigations picker as pathology — no change for this step.
- 2

Reception books a machine slot
RECEPTION
Picks a machine/room and an available time slot instead of a collection date/window.
- 3

Radiographer checks in, performs, drafts findings
RADIOGRAPHER

Consent captured at check-in if required; performing the study and writing an initial findings/impression draft happen in one worklist step.

4 Radiologist authors, then verifies & releases

RADIOLOGIST

Edits/finalizes the draft, then verifies — posts the bill charge and releases the report in one action.

3.2 Outsourced Labs

1 Collected exactly like a pathology sample

PHLEBOTOMIST

Same Collection Worklist, same accession-number mechanics — a blood draw is a blood draw regardless of which lab processes it.

2 Received, then batched into a dispatch

LAB TECHNICIAN

Multiple items going to the same partner lab are grouped into one dispatch manifest.

3 Items return independently

LAB TECHNICIAN

Partial returns are tracked — a 3-item dispatch can come back as 1, then 2 more, later.

4 Result entry & verification — same screens as pathology

LAB TECHNICIAN PATHOLOGIST

Once returned, values are keyed in and verified through the exact same Result Entry / Verification screens already used for in-house tests.

4. Business Rules Reference (new in Slice 2)

RULE	WHAT IT ENFORCES
BR-14	Two bookings can never win the same machine/time slot — enforced at the database level, not just the UI.
BR-16	A test priced at or above a configured threshold on a TPA/Corporate rate list is held pending insurance pre-authorization before it can proceed.
BR-03 (extended)	Whoever authors imaging findings cannot also verify them — same rule as pathology, now also covering Imaging, with a per-tenant off switch for a single-radiologist hospital.
BR-10 (imaging exception)	A Pathology test needs at least one parameter before it can be activated; an Imaging test does not (it uses free-text findings, not parameters) — confirm this distinction specifically, see TC-NEG-01.

5. Role-Based Test Cases

ADMIN — Machines & Resources

Login: ADMIN-001 / Admin@12345

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC-RES-01	Diagnostics Machines & Resources "Add Machine/Room." Create e.g. "MRI Machine 1," set working hours (e.g. Mon–Sat 08:00–20:00).	Machine appears in the list, Active by default. Working hours are saved and listed on the detail panel.	
TC-RES-02	Add a second machine of the same type (e.g. "MRI Machine 2"). Add a blackout date on Machine 1 covering tomorrow (e.g. "annual maintenance").	Both machines are independently manageable. Machine 1 shows no available slots tomorrow; Machine 2 is unaffected.	
TC-RES-03	Deactivate a machine.	It no longer appears as a bookable option when booking an imaging test.	

ADMIN — Pre-Authorization Thresholds

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC-PRE-01	Diagnostics Pre-Auth Thresholds. Set a threshold of 5,000 for Imaging.	Threshold is saved and listed. Setting it again for the same test type updates it in place rather than creating a duplicate row.	
TC-PRE-02	Book an imaging test priced above 5,000 on a TPA rate list (not Cash).	The order item is created with pre-authorization PENDING . Attempting to check in / collect / dispatch it is blocked with a clear pre-auth message until a pre-auth number is recorded.	
TC-PRE-03	Book the same test on a Cash rate list instead.	No pre-authorization is required — the gate only applies to TPA/Corporate payer types.	

RECEPTION — Booking an Imaging Slot

Login: MBSRECP2026001 / Test@12345

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC-BOOK-01	Precondition: a patient has a pending Imaging recommendation. Open Book Tests, select it, choose a machine and an available time slot, book.	Order is created; the slot picker only ever shows genuinely free times (booked/blacked-out times are absent or disabled).	

TC-BOOK-02	Select both a pending Pathology test and a pending Imaging test at once for the same patient.	Only one test type is bookable per trip to the counter — selecting one greys out the other type, with a message explaining they must be booked separately.
TC-BOOK-03 (needs two browser sessions/tabs)	Two reception staff attempt to book the same machine, same date, same time slot for two different patients at essentially the same moment.	Exactly one booking succeeds. The other gets a clear "that slot was just taken — pick another" message, not a crash or a silent double-booking.

RADIOGRAPHER — Imaging Worklist

Login: MBSRADIOG001 / Test@12345

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC-IMG-01	Precondition: a study requiring consent is booked. Attempt "Check in" without ticking the consent checkbox.	Blocked with a clear message that consent is required first.	
TC-IMG-02	Tick consent, check in, then "Start study," then "Complete study" — enter Technique/Findings/Impression.	Study moves through each stage visibly; completing it sends it to the Radiologist's Imaging Verification queue.	

RADIOLOGIST — Imaging Verification

Login: MBSRADIOL001 and MBSRADIOL002 / Test@12345

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC-RVIFY-01	As MBSRADIOL001, open the drafted study, edit the impression if needed, click "Save" (authors it). Then, still as the same login, attempt "Verify & Release."	Blocked — "the radiologist who authored this study cannot also verify it." This is the default, enforced behavior.	
TC-RVIFY-02	Log in as MBSRADIOL002 instead and verify the same study.	Succeeds. Bill charge posts, PDF report releases with Technique/Findings/Impression (not a parameter table — that layout is pathology-only).	
TC-RVIFY-03 (Admin-only setup)	As Admin, open Super Admin this tenant turn off "Require Separate Verifier" for Imaging. Repeat TC-RVIFY-01.	The same login can now author <i>and</i> verify. Turn the toggle back on afterward to restore the default.	

Login: MBSLABTECH001 / Test@12345

ID	PRECONDITION & STEPS	EXPECTED RESULT	PASS/FAIL
TC-OUT-01	Precondition: an outsourced order has been collected and received (same flow as pathology). Open Outsourced Labs Dispatch tab, select 2+ items for the same partner lab, dispatch.	One dispatch manifest is created covering all selected items; each item's status moves to "Dispatched."	
TC-OUT-02	Go to the Returns tab. Mark just one of the two dispatched items as returned.	The dispatch shows "Partial Return" — the other item stays "Dispatched," not silently closed out.	
TC-OUT-03	Mark the second item returned too.	Dispatch flips to fully "Returned." The returned item(s) now appear on the normal Result Entry worklist, exactly like a pathology sample.	
TC-OUT-04	Enter a result value on the returned item, submit, then verify as a Pathologist (MBSPATH001).	Same verify screen as pathology; charge posts and report releases identically — the hospital's own pathologist signs off even though a partner performed the analysis.	
TC-OUT-05	Set a past date as a dispatch's "expected return" date (or wait for a real one to lapse) and check the Returns tab.	That dispatch is visibly flagged overdue.	

6. Smaller Additions

ID	SCENARIO	EXPECTED RESULT	PASS/FAIL
TC-MISC-01	Diagnostics Pending Follow-Up (Reception). A recommendation has sat unbooked for 3+ days.	It appears in this list with how many days it's been pending. "Book tests" and "Going outside" (decline) actions both work from here.	
TC-MISC-02	Open an order that was booked with partial payment (e.g. from the "View order" link after booking). Add the remaining balance.	Balance updates to 0, payment status flips to "Paid."	
TC-MISC-03	Log in as Reception and open the Collection Worklist page directly.	Page now loads and shows the due-sample list (previously blocked). Reception still cannot click "Collect" themselves — that action correctly stays restricted to Phlebotomist/Lab Technician.	

7. Cross-Cutting & Negative Test Cases

ID	SCENARIO	EXPECTED RESULT	PASS/FAIL
TC-NEG-01	As Admin, create a new Imaging test and activate it without adding any parameters.	Activation succeeds — this is correct, not a bug. Imaging tests use free-text findings, not parameters, so the "must have at least one parameter" rule only applies to Pathology tests.	
TC-NEG-02	As a Lab Technician (not a Pathologist), attempt to verify any result — pathology or outsourced.	Blocked by role, regardless of who entered it — verification is Pathologist-only.	
TC-NEG-03	Confirm existing Slice 1 pathology flows (recommend book collect receive enter verify bill report) still work exactly as before.	No regression — this build only adds new capability, it does not change pathology's existing behavior.	

8. Known Limitations / Out of Scope

- **Imaging is report-only** — no DICOM image storage or in-app viewer. Actual X-ray/CT/MRI images stay on the hospital's own imaging equipment/PACS; the HMS tracks the order, workflow, and written report only.
- **No PACS/analyser integration** — results are entered manually; the system does not receive data automatically from lab or imaging machines.
- **Consent capture is a simple checkbox**, not a legally binding e-signature system.
- **Staff-to-machine rostering** (which radiographer is assigned to which machine, by shift) is not built — any Radiographer can act on any machine's worklist today.

- **Mixed-type single order** — a patient with both a pending pathology and a pending imaging test must be booked in two separate trips to the counter, not one combined order (see TC-BOOK-02).